

MUHAMMAD DANISH

AI Engineer | Machine Learning Engineer | Generative AI Application Developer

+92 309 9598241 | mianmuhammaddanishkhattak@gmail.com | linkedin.com/in/muhammad-danish-khattak | github.com/MianDanish1122 | Lahore, Pakistan

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building real-world AI solutions across Machine Learning, Deep Learning, Computer Vision, and Generative AI. Currently working on voice AI agents, computer-vision-based interior styling systems, and practical AI automation to solve real business and consumer problems. Experienced integrating AI models into production-ready applications using Python, TensorFlow, PyTorch, Scikit-learn, OpenCV, and FastAPI. Runner-Up at ByteSprint 2025 and an active participant in several national and international hackathons, with additional entrepreneurial experience running an e-commerce business.

EDUCATION

Bachelor of Computer Science — Specialization in Artificial Intelligence | Abdul Wali Khan University Mardan, Pakistan | Sept 2022 – 2026

Courses: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Data Engineering, Web Development, Database Systems, Software Engineering.

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Codecilex, Pakistan

- Worked on computer vision and AI automation initiatives, integrating AI models into practical, production-ready applications.
- Collaborated with cross-functional teams to deliver production-ready AI solutions.

Text-to-Speech (TTS) Voice Agent [Python | TTS Frameworks | FastAPI]

- Developed a TTS voice agent that converts text input into natural, real-time speech output for automated speech applications.
- Designed the pipeline for conversational use cases, enabling automated, hands-free voice interaction.
- Integrated the agent via a FastAPI backend to support easy deployment into other applications.

AI Room Scan & Interior Styling System [Python | OpenCV | Generative AI | FastAPI]

- Built a computer-vision pipeline that scans a room image and identifies layout, furniture, and empty space.
- Combined the vision output with generative AI to produce personalized interior styling recommendations.
- Exposed the system through a FastAPI backend for integration into a practical, end-user-facing application.

PROJECTS

Analyzing Student Behavior Before and After Classroom Sessions Using Deep Learning [Python | TensorFlow/PyTorch | OpenCV | Deep Learning | FER2013]

- Built a computer-vision system to analyze students' facial emotions before and after classroom sessions using the FER2013 dataset.
- Compared pre- and post-lecture emotional states to measure engagement and attention trends for instructors.

AarogyaNet — Multi-Agent Healthcare Assistant [Python | LLMs | LangChain | LangGraph | RAG | FastAPI | Streamlit]

- Designed a multi-agent healthcare assistant with specialized agents for symptom analysis, medical knowledge retrieval, and wellness recommendations.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline over a vector database to ground responses in trusted medical information.
- Built a coordinator agent to route requests and combine outputs from specialized agents into a single structured response.

AI-Social-Agent [Python | LLMs | Generative AI | Prompt Engineering | LangChain | Streamlit]

- Built an AI content assistant that generates and rewrites social media posts, captions, and hashtags from simple prompts.
- Applied prompt engineering to adapt content tone and formatting across LinkedIn, Instagram, and X.

AI Career System [Python | Scikit-learn | Pandas | NumPy | Streamlit]

- Built an AI-powered career recommendation platform that analyzes user skills, education, and interests to suggest suitable career paths.
- Implemented skill-gap analysis and personalized course/certification suggestions using classification and recommendation techniques.

Flood AI Chatbot [Python | NLP | Prompt Engineering]

- Built a conversational chatbot during the Flood.pk Hackathon (NIC Peshawar) to help users get flood-related information.
- Applied prompt engineering to keep responses accurate, relevant, and easy to understand for the general public.

Humrahi Hub [Web Platform | Blockchain-Inspired Transparency — Hackathon Winning Project]

- Built a community-driven platform to solve the "Someone Paradox" by connecting verified community needs with donors and volunteers.
- Designed blockchain-inspired transparency and automated fund-flow tracking so every donation is visible and accountable.
- Won 1st Position at the 2nd AWKUM AGORA 2026 Hackathon as part of Team Tec-Hub.

Fake News Detection *[Python | Scikit-learn | NLP]*

- Trained an NLP-based text classifier to distinguish between reliable and unreliable news articles.
- Applied text preprocessing and feature extraction techniques to improve classification quality.

Heart Disease Predictor & Cancer Stage Prediction *[Python | Scikit-learn | Pandas | NumPy]*

- Built two separate ML classification models to predict heart disease risk and cancer stage from patient health data.
- Performed data cleaning, feature engineering, and model evaluation to improve prediction reliability.

Credit Risk Analysis & Email Spam Detection *[Python | Scikit-learn | Feature Engineering]*

- Developed a credit risk classification model to support financial risk scoring decisions.
- Built a separate email spam detection model using text-based feature engineering and classification techniques.

Movie Review & Financial News Sentiment Analysis *[Python | NLP | Scikit-learn]*

- Built an NLP sentiment classifier to analyze and label movie reviews as positive or negative.
- Built a second sentiment analysis model focused on financial news to gauge market-related sentiment.

Lung Disease Detection using Lung Sounds *[Python | TensorFlow/Keras | Deep Learning — University Project]*

- Applied audio signal processing to convert lung sound recordings into model-ready features.
- Trained a deep learning model to classify lung sounds and detect signs of lung disease.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning & Deep Learning: Scikit-Learn, TensorFlow, PyTorch, Neural Networks, Transfer Learning, Model Optimization

Generative AI & LLMs: LLMs, RAG Systems, LangChain, LangGraph, Prompt Engineering, Agentic AI, Multi-Agent Orchestration, OpenAI API

NLP & Transformers: Natural Language Processing, Hugging Face Transformers, Text Processing, Embeddings, Semantic Search

Computer Vision: YOLOv11, Object Detection, Image Segmentation, VGG16, CNN

Vector Databases & Search: FAISS, ChromaDB, Qdrant, Hybrid Retrieval

Backend & APIs: FastAPI, REST APIs, API Integration, Streamlit, Gradio, Next.js

MLOps & Deployment: Docker, MLflow, CI/CD Pipelines, Git, GitHub, Model Deployment

Data Analysis: Pandas, NumPy, Feature Engineering, Data Preprocessing, EDA

CERTIFICATIONS & TRAINING

- Machine Learning, AI & Data Science — eHunar
- Generative AI Application Developer — NCEAC HEC
- KPITB AI/ML Program
- Machine Learning — Coursera
- Python — Coursera
- Multiple additional Coursera certificates in AI, ML, and Python (6–8 certificates)

AWARDS & ACHIEVEMENTS

- 1st Position — 2nd AWKUM AGORA 2026 Hackathon (Team Tec-Hub, Project: Humrahi Hub)
- Runner-Up — ByteSprint 2025 Speed Coding Competition
- Participant — NeoHacks GIKI Edition 2025
- Participant — Flood.pk Hackathon (NIC Peshawar)
- Participant — IUPC 2026
- Participant — Hack Nation Global AI Hackathon
- Participant — Studiio Hacks

LANGUAGES

English (Professional Working Proficiency) | Urdu (Fluent) | Pashto (Native)